The Temporal Blind Spot in Video Retrieval: Diagnosing Temporal Sensitivity

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Abstract

Temporal blindness in video models is well-documented for generative QA, but its consequences for *retrieval embeddings*—where failures are silent—remain uncharacterized. We diagnose temporal sensitivity across six retrieval benchmarks spanning copy detection, temporal manipulation, and high-overlap motion retrieval. A feature-vs-comparator decomposition reveals that, within a given encoder, the matching stage (per-frame representation and sequence-aware similarity) governs temporal sensitivity: on Honda HDD this switch explains 89% of the performance gap between bag-of-tokens and the temporal residual (point estimate; cluster-level block-bootstrap CIs are wider), with a non-DTW set-matching aggregation already closing 76%; on nuScenes the matching stage alone is on par with the learned residual transform (AP = 0.867 vs. 0.815; CIs marginally overlap). A *temporal scramble gradient* ($K \in \{1, \ldots, 16\}$) cleanly separates order-invariant from sequence-aware methods. Embedding probes across three open-weight Video-LLMs localize where temporal signal is lost: temporal-derivative DTW separates forward from reverse perfectly—an algebraic property of derivatives under reversal, not of the features themselves—but mean-pooling erases this signal ($s_{\text{rev}} \approx 1.0$). Generative probes on Claude 4.6 Opus and Gemini 3.1 Pro fare no better than the open-weight models. Neither linear nor nonlinear probes recover temporal order from LLM hidden states (126 configurations, though the $p \gg n$ regime bounds recoverability rather than proving absence). Across all evaluated methods, no single approach spans both copy detection and motion retrieval, exposing a sensitivity–invariance trade-off that current architectures do not resolve. We release open-source code packaging the scramble gradient and reversal tests.

1 Introduction

Modern video retrieval represents videos as mean-pooled frame embeddings from pretrained encoders. This *bag-of-frames* (BoF) approach excels at Video Copy Detection (VCD)—identifying re-encoded, cropped, or camcorded copies—because visual semantics survive frame reordering, enabling billion-scale nearest-neighbor search [21, 20].

BoF discards temporal order entirely. A growing literature documents this "temporal blindness" in Video-LLMs [45, 46, 42, 52, 53, 50]: generative probes show near-chance temporal QA; frame shuffling leaves benchmark scores intact. These diagnoses target generative QA, where failures surface in text. In retrieval, failures are silent—reversed videos return identical embeddings, left turns match right turns—and no downstream signal flags them. We fill this gap with three test cases:

1. Copy Detection & Scene Retrieval: when the discriminative signal is visual background (VCD; activities in different rooms), semantic pooling suffices.

Preprint.

2. Temporal Manipulation (Trust & Safety): reversed video yields BoF similarity of 1.0, evading embedding-based copy retrieval [8].

3. High-Overlap Motion Retrieval (Dashcam Fleets): querying a fleet for "left turns at Intersection X" requires encoding motion dynamics; static background dominates the BoF embedding, conflating left and right turns.

We evaluate these cases using DINOv3 [3] (BoF and DTW attention trajectories), V-JEPA 2 [1] (spatiotemporal prediction residuals), ViCLIP [58] (video-native contrastive baseline), TARA [36] (chiral-trained MLLM), PL-Stitch [10] (temporal ranking pretraining), and three Video-LLMs (Qwen3-VL [14], Gemma 4 [15], LLaVA-Video [16]), across six benchmarks: VCDB [4], Nymeria [32], MUVR [19], Honda HDD [5], nuScenes [6], and EPIC-Kitchens-100 [7].

Contributions. We contribute a diagnostic methodology for temporal sensitivity in video retrieval: (1) a continuous *temporal scramble gradient* separating order-invariant from sequence-aware methods (Figure 2); (2) a *feature-vs-comparator decomposition* with VLM-pipeline probes that localize where order signal dies, extending temporal-blindness diagnosis from generative QA to retrieval; (3) a sampling-density analysis on Honda HDD; (4) open-source tooling packaging the scramble gradient and reversal tests. Applied across six benchmarks, the suite surfaces a sensitivity–invariance trade-off current architectures do not resolve.

2 Methods

Symmetric aggregation implies permutation-invariance. Mean-pooling, max-pooling, and Chamfer matching are symmetric aggregators: they satisfy $g(f_1, \ldots, f_T) = g(f_{\pi(1)}, \ldots, f_{\pi(T)})$ for all permutations π [60]. BoF order-invariance therefore follows from the aggregation operator, independent of encoder capacity or training objective.

DINOv3 & Attention Trajectories. We extract 1024-d CLS embeddings per frame from DINOv3 ViT-L [3]. *BoF* mean-pools frame embeddings into a single vector, compared via cosine similarity. *Chamfer similarity* [26] computes the pairwise frame similarity matrix and averages the symmetric maximum: $\frac{1}{2}\big(\mathrm{mean}_i \max_j S_{ij} + \mathrm{mean}_j \max_i S_{ij}\big)$, remaining order-invariant while retaining per-frame structure. For structural tracking, we compute the spatial center-of-mass of the CLS→patch attention map (last layer, averaged across heads) per frame, yielding a 2D trajectory compared via DTW. Temporal derivatives of CLS embeddings capture frame-to-frame transition dynamics, also compared via DTW.

V-JEPA 2 Temporal Residuals. V-JEPA 2 ViT-L [1] (see also V-JEPA 2.1 [2]) ingests 64 frames as $32 \times 16 \times 16$ spatiotemporal patches ($D = 1024$). The *prediction residual* $R = \hat{P} - P$ (predicted minus actual masked patches) captures motion-specific error patterns. We spatially average each temporal position to obtain $\bar{R} \in \mathbb{R}^{T_{\text{mask}} \times D}$, compared via DTW with similarity $\exp(-d_{\text{DTW}})$. The *bag-of-tokens* (BoT) baseline mean-pools all encoder patches into a single vector (cosine similarity).

Temporal Scramble Gradient. We split video B's frame embeddings into K equal chunks, randomly shuffle them, and reassemble. Sweeping $K \in \{1, \ldots, 16\}$ traces a gradient from intact to fully scrambled. We shuffle *extracted* embeddings (see Appendix O). Order-invariant methods should remain flat; sequence-aware methods should degrade.

VLM Generative Probes. We present forward and reversed clips to VLMs, classify temporal direction with three prompts, and report balanced accuracy.

ViCLIP. ViCLIP ViT-L [58], trained contrastively on 10M video clips, jointly encodes 8 frames via temporal attention into a single 768-d embedding (cosine similarity).

TARA. TARA [36] fine-tunes a Tarsier-7B MLLM with chiral (reversed-video) negatives for temporal retrieval, encoding 16 frames into a single 4096-d vector via end-of-line prompting (cosine similarity).

PL-Stitch. PL-Stitch [10] pretrains a ViT-Base (86M params) with Plackett-Luce temporal ranking, producing per-frame 768-d CLS-token embeddings that support the same feature-vs-comparator decomposition as DINOv3: BoF, Chamfer, temporal derivative DTW, and DTW on raw embeddings.

VLM Embedding Probes. We extract three intermediate representations from VLM pipelines: (1) *vision pooled*—mean-pooled vision tower output (BoF over frame×patch tokens); (2) *vision*

sequence—per-frame pooled vision tokens as a (T, D) sequence, compared via temporal derivative DTW; (3) *LLM hidden state*—mean-pooled last-layer hidden state over all input tokens. We measure s_{rev}, the similarity between forward and reversed representations of the same clip:

$$s_{\text{rev}} = \begin{cases} \cos(\mathbf{z}_{\text{fwd}},\ \mathbf{z}_{\text{rev}}) & \text{pooled probes (single vector),} \\ \exp\big(-\alpha \cdot d_{\text{DTW}}(\mathbf{Z}_{\text{fwd}},\ \mathbf{Z}_{\text{rev}})\big) & \text{sequence probes (temporal derivative DTW),} \end{cases} \tag{1}$$

where α follows the DTW α schedule defined below. $s_{\text{rev}} \approx 1.0$ indicates order-invariance; lower values indicate order sensitivity. Raw s_{rev} values exceeding 1.0 (from f32 cosine precision) are clamped to 1.000 in all tables and figures; affected entries are marked †. Because the two cases use different similarity functions, pooled and sequence s_{rev} values are not directly comparable (see Discussion).

DTW Implementation. All DTW comparisons use unconstrained warping with L2 cost matrices, path-length normalization ($d/(T_1+T_2)$), and GPU-vectorized anti-diagonal wavefront processing [22]. Similarity is $\exp(-\alpha \cdot d_{\text{DTW}})$ with $\alpha = 1$ for derivatives/residuals, $\alpha = 5$ for attention trajectories (min-max normalized to $[0, 1]$). Because $\exp(-\alpha \cdot d)$ is monotone in d, rankings are α-invariant; we verified this across three representative cells with $\alpha \in \{0.5, 1, 2, 5, 10\}$ (max AP spread $< 10^{-3}$).

3 Experiments and Results

Evaluation protocols. All evaluations are pairwise. *VCDB*: 528 videos; 5,585 positive pairs from annotations, negatives sampled 1:1 with fixed seed (sensitivity analysis across 1:1, 1:5, and all-pairs in Appendix I). *HDD*: all $\binom{n}{2}$ pairs within each GPS-clustered intersection (DBSCAN, $\epsilon \approx 30$ m); positive = same maneuver; 50 mixed clusters, 1,687 maneuvers. *nuScenes*: same protocol on ego_pose; CAN bus–derived labels; 264 segments, 950 pairs. *EPIC* s_{rev}: per-sequence self-comparison on 500 sequences; bootstrap 95% CIs (1,000 resamples). *Nymeria*: 100 egocentric sessions, 8,780 segments; positive = same activity same room; all $\binom{n}{2}$ pairs (394,210). *MUVR News*: 9,958 videos, 74 topics, 27,453 positive pairs.

3.1 VCDB Temporal Manipulation Diagnostics

On copy detection, BoF and Chamfer dominate: DINOv3 Chamfer reaches AP = 0.989 (528 videos, 5,585 positive pairs); temporal methods underperform on frame-to-frame intra-class noise. Individual VCDB AP values swing by ± 0.06 across random negative seeds (Appendix I), but rank order is preserved. Nymeria and MUVR replicate the ranking (Appendix D).

Order-invariance is also a vulnerability: reversed video yields identical BoF similarity. We measure this across VCDB positive pairs:

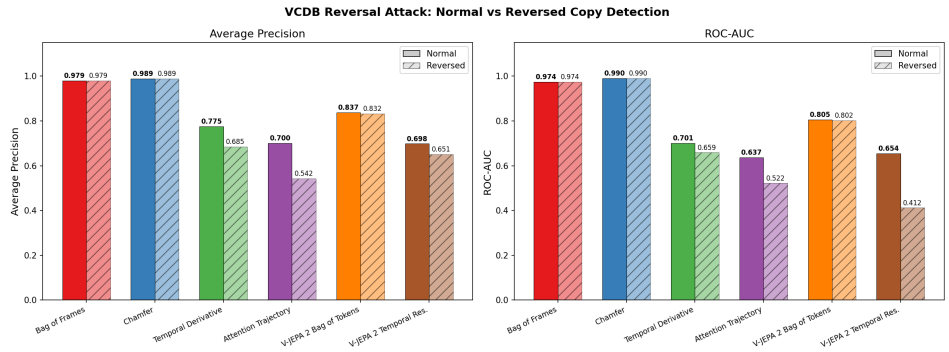

Figure 1: VCDB Reversal Attack (528 videos, 5,585 positive pairs). Order-invariant methods (BoF, Chamfer) maintain identical AP/AUC under reversal. Only sequence-aware methods (Attention Trajectory) register the structural tampering via an AP drop.

BoF and Chamfer yield zero AP delta under reversal. Only attention-trajectory DTW flags the reversed copy ($\Delta = -0.158$); the temporal residual registers a smaller drop ($\Delta = -0.067$). To

Table 1: VCDB Reversal Attack: 5,585 positive pairs (AP drop indicates detection of evasion).

Method	**Normal AP**	**Reversed AP**	**AP Δ**
Chamfer (DINOv3)	0.989	0.989	**-0.000** (Invariant)
BoF (DINOv3)	0.979	0.979	**-0.000** (Invariant)
BoT (V-JEPA 2)	0.838	0.832	-0.006
Attention Trajectory (DINOv3)	0.700	0.542	**-0.158** (Detects)
Temporal Residual (V-JEPA 2)	0.698	0.631	-0.067

move beyond this binary test, we apply chunk-shuffle scrambling at varying granularity ($K \in \{1, \ldots, 16\}$).

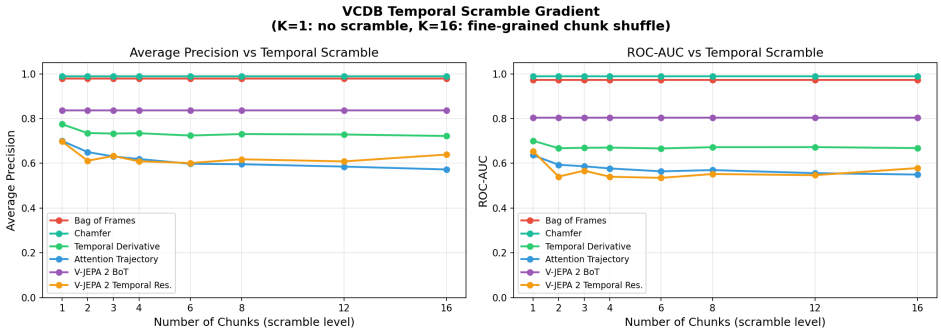

Figure 2: Temporal Scramble Gradient on VCDB (527 of 528 videos; one decode failure; 5,585 positive pairs). Order-invariant methods (BoF, Chamfer, V-JEPA 2 BoT) maintain constant AP regardless of scramble severity. Sequence-aware methods show overall degradation, with attention trajectories declining most steeply (-18.3% relative); V-JEPA 2 residual exhibits non-monotonic variation, possibly due to boundary effects at chunk boundaries (Appendix O).

The gradient separates method classes (Figure 2; Appendix O): order-invariant methods stay flat; attention trajectories degrade most (-18.3% relative from $K = 1$ to $K = 16$), consistent with gaze paths carrying the most temporal structure.

3.2 High-Overlap Motion: Honda HDD & nuScenes

To isolate motion from appearance, we clustered 128 Honda HDD [5] driving sessions by RTK GPS, extracting 1,687 maneuvers at 50 mixed-direction intersections where BoF similarity averages 0.63 ± 0.18 regardless of turn direction (Appendix L). The full dataset is passage-dominated ($\sim 66\%$), so AP is partially inflated by easier passing-vs-turn negatives.

DINOv3 methods fail (AP $\approx$ 0.50): dashcam attention tracks salient objects identical for both turns. V-JEPA 2 temporal residuals achieve AP = 0.956; a pixel-difference motion proxy rules out a trivial intensity shortcut (Spearman $r = 0.051$; Appendix H), and optical flow (RAFT) achieves only AP = 0.478. Restricting to left-vs-right pairs yields AP = 0.960 [0.958, 0.963]. Comparing encoder patches via DTW (encoder-sequence DTW) achieves AP = 0.942, showing that the matching stage (pooled-cosine $\rightarrow$ per-frame DTW) explains 89% of the BoT-to-residual gap (0.825 $\rightarrow$ 0.956). A pairwise *symmetric-max* (Chamfer) aggregation [26] on the same V-JEPA 2 encoder-sequence patches further decomposes this: it reaches AP = 0.914 without any dynamic programming, closing 76% of the BoT$\rightarrow$Encoder-Seq-DTW gap. The remaining 24% (0.914$\rightarrow$0.942) is attributable specifically to DTW's monotonic warping constraint. We restrict encoder-sequence DTW to motion tasks (HDD, nuScenes); on VCDB, BoF already hits AP = 0.989.

TARA [36], a 7B MLLM trained with chiral negatives, hits AP = 0.547 under cosine—indistinguishable from ViCLIP (0.542) despite $\sim 23\times$ more parameters. PL-Stitch [10] (ViT-Base, Plackett-Luce ranking) supports the same decomposition: BoF yields AP = 0.478, DTW on raw embeddings 0.540—both near chance, and below DINOv3 Chamfer (0.559), possibly reflecting a capacity (86M vs. 300M) and pretraining-scale gap rather than a comparator effect. Yet $s_{\text{rev}} = 0.006$

Table 2: Honda HDD Maneuver Discrimination: 1,687 maneuvers across 128 sessions, 50 mixed-direction intersection clusters (same vs. different turn at same intersection).

Method	**AP**	**AUC**
DINOv3 Temporal Deriv.	0.498	0.526
DINOv3 Attn. Trajectory	0.507	0.541
DINOv3 BoF	0.530	0.542
DINOv3 Chamfer	0.559	0.577
ViCLIP (InternVid-10M)	0.542	0.549
TARA (chiral, cosine)	0.547	0.600
PL-Stitch BoF	0.478	0.512
PL-Stitch Chamfer	0.498	0.547
PL-Stitch Temporal Deriv.	0.492	0.523
PL-Stitch DTW (raw emb.)	0.540	0.548
V-JEPA 2 BoT	0.825	0.836
V-JEPA 2 Chamfer	0.914	0.912
V-JEPA 2 Encoder-Seq DTW	0.942	0.935
V-JEPA 2 Temporal Residual	**0.956**	**0.952**

under raw DTW confirms PL-Stitch *does* encode frame order—the features are not discriminative enough for maneuver retrieval.

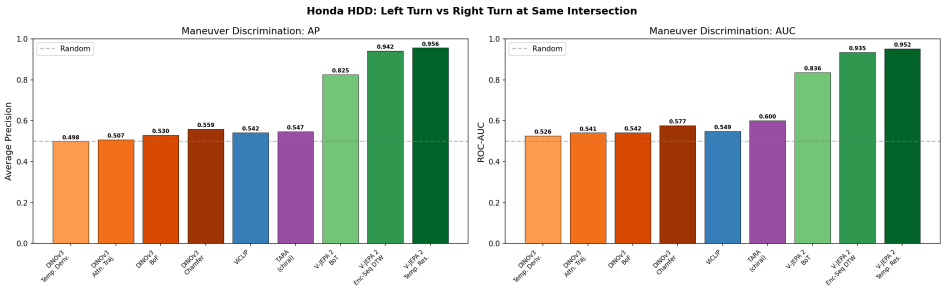

Figure 3: Honda HDD maneuver discrimination (1,687 maneuvers, 128 sessions). V-JEPA 2 temporal residuals achieve near-perfect separation (AP = 0.956) while DINOv3 methods remain at chance.

3.2.1 Sampling Density: Context Window Sweep

We fix the backbone and pair set on HDD and sweep clip duration $d \in \{1.0, \ldots, 8.0\}$ seconds, always extracting 64 unique frames ($\text{fps}_{\text{eff}} = 64/d$). This avoids the padding confound of the FPS downsample experiment (Appendix A).

3.2.2 Cross-Dataset Validation: nuScenes

To test whether the ranking transfers, we evaluate on nuScenes [6]. Turn labels come from CAN bus; intersections are clustered via DBSCAN ($\epsilon = 30$m), yielding 264 segments across 50 mixed-direction clusters (950 pairs).

The ranking is directionally consistent (Table 5): V-JEPA 2 encoder-sequence DTW reaches AP = 0.867, directionally ahead of the temporal residual (0.815) and BoT (0.791); DINOv3 methods stay $\leq$0.623. CIs overlap marginally between the top two ($[0.842, 0.890]$ vs. $[0.785, 0.844]$; Appendix J), so the residual-vs-DTW ordering is suggestive rather than significant; the BoT gap is definitive. Even without the learned residual, the matching stage is the dominant driver. The residual helps on HDD but not nuScenes, likely because nuScenes' native 2 Hz yields sparser temporal gradients where prediction errors add noise; however, nuScenes at 2 Hz still beats HDD subsampled to 2 FPS with repeat-padding (AP = 0.636), suggesting native keyframes preserve more temporal signal (Figure 8, Appendix M).

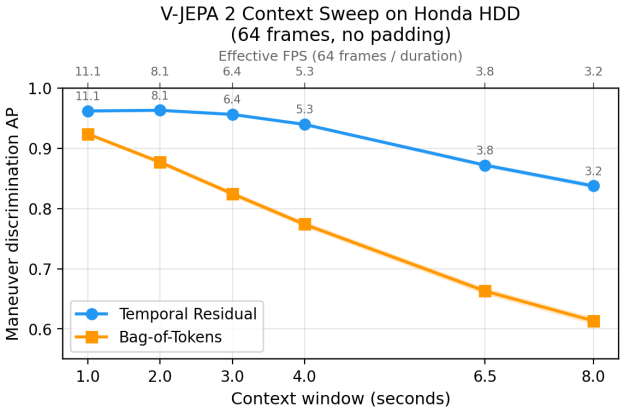

Figure 4: V-JEPA 2 context sweep on Honda HDD (1,687 maneuvers, 128 sessions). Temporal residual degrades more slowly than BoT; the gap grows from $+0.038$ at 1 s to $+0.224$ at 8 s (full table in Appendix B).

3.3 VLM Temporal Blindness (EPIC-Kitchens)

Prior work documents temporal blindness in Video-LLMs for generative QA [45, 46, 42, 52, 53]. We extend this to *retrieval embeddings*, probing where temporal signal dies. We evaluate three VLMs on 500 EPIC-Kitchens-100 [7] sequences: Qwen3-VL-8B, Gemma-4-31B, and LLaVA-Video-7B.

3.3.1 Generative Probes

Each model received identical frames with three semantically equivalent prompts asking to classify temporal direction.

Table 3: VLM forward/reverse classification on EPIC-Kitchens (500 sequences for open-weight models, 200 for API models). All at or near chance. $\pm$ values are standard deviation across prompts (not statistical uncertainty); binomial 95% CIs at $N = 200$ are approximately ± 0.035.

Model	**Prompt 0**	**Prompt 1**	**Prompt 2**	**Mean Bal. Acc.**
Qwen3-VL-8B	0.517	0.502	0.504	0.508 ± 0.007
Gemma-4-31B	0.549	0.528	0.535	0.538 ± 0.009
LLaVA-Video-7B	0.503	0.501	0.505	0.503 ± 0.002
Claude 4.6 Opus [54]	0.490	0.590	0.555	0.545 ± 0.055
Gemini 3.1 Pro	0.585	0.608	0.603	0.599 ± 0.010

All five models perform at or near chance; Gemini 3.1 Pro [55] (a reasoning model) reaches 0.599 ± 0.010—marginally above chance but far from reliable. Answers are driven by prompt phrasing, not temporal direction (Appendix F and G).

3.3.2 Embedding Probes: Where Temporal Signal Dies

VLM *outputs* carry no temporal information. We probe whether temporal signal persists in internal representations before being discarded.

Order signal is recoverable from per-frame features but erased by pooling. Pooled VLM vision embeddings are order-invariant ($s_{\text{rev}} \approx 1.0$). Retaining per-frame tokens and comparing via temporal derivative DTW achieves perfect forward/reverse separation (balanced accuracy = 1.000) for both CLIP and SigLIP [38]. This follows from the mathematical structure of temporal derivatives under DTW (reversed derivatives are negated), not learned understanding. Qwen3-VL's pooled vision embeddings underperform sharply (VCDB AP = 0.491 vs. 0.915–0.926), likely because its 3D temporal–spatial merging entangles frame identity with spatial position—a limitation inherent to the 3D-merging architectural class rather than a model-specific failure.

Table 4: Order sensitivity (s_{rev}) across the VLM pipeline on EPIC-Kitchens (500 sequences) with bootstrap 95% CIs where available. $s_{\text{rev}} \approx 1.0$ = order-invariant; lower = more order-sensitive. DINOv3 structural fingerprints shown for reference.*

Method	$s_{\text{rev}}\downarrow$ **[95% CI]**	**Bal. Acc.**$\uparrow$
DINOv3 structural fingerprints (reference)		
BoF	1.000	N/A
Chamfer	0.9999	N/A
Temporal Derivative	0.500 [0.499, 0.500]	1.000
Attention Trajectory	0.192 [0.187, 0.197]	1.000
V-JEPA 2		
BoT	0.984	0.500
Temporal Residual	0.699	0.778
VLM vision towers (pooled)		
Gemma-4 (SigLIP)	1.000†	N/A
LLaVA (CLIP)	1.000†	N/A
VLM vision sequence (per-frame DTW)		
Gemma-4 (SigLIP)	0.492 [0.491, 0.493]	1.000
LLaVA (CLIP)	0.495 [0.495, 0.496]	1.000
VLM LLM hidden state		
Gemma-4 (Gemma)	1.000 [1.000, 1.000]	N/A
LLaVA (Qwen2)	0.998 [0.998, 0.999]	N/A
Qwen3-VL (Qwen3)	0.998 [0.998, 0.999]	N/A

*DINOv3 300M params, CLIP (ViT-L) 304M, SigLIP (SO-400M) 400M. Order-invariance follows from the aggregation operator [60], not model capacity. †Raw values 1.001–1.002 from f32 precision; clamped to 1.000. Qwen3-VL vision probes are omitted from this table because its ViT uses dynamic-resolution spatial grids with 3D temporal–spatial merging, which disrupts clean per-frame token grouping for the detailed s_{rev} analysis; the mechanism behind its anomalously low pooled VCDB AP (0.491 vs. 0.915–0.926 for CLIP/SigLIP; Table 6) warrants further investigation. Approximate pooled and sequence results appear in Table 6. The LLM hidden-state probe (below) is unaffected.

The LLM backbone does not surface this signal under any pooled readout ($s_{\text{rev}} \approx 1.0$ across all layers; Appendix C). Neither logistic regression nor 2-layer MLP probes on per-position LLM hidden states recover temporal order above chance (best: 0.560; all 126 configurations within chance range; Appendix P). Yet per-position hidden states *do* retain order signal under full-sequence DTW (Appendix N). Symmetric aggregation, not the LLM backbone itself, is the bottleneck.

3.4 Cross-Method Summary

Tables 5 and 6 consolidate results across all benchmarks; no single method dominates every paradigm. Table 5 covers dedicated video/image encoders used directly for retrieval; Table 6 covers the VLM pipeline (generative classification and embedding probes).

4 Related Work

Video copy detection. Standard benchmarks (ISC [23], VSC [24], VCSL [28]) and descriptors (SSCD [25], ViSiL [26], DnS [27], S2VS [47]) provide per-frame features with learned aggregation. Wang et al. [51] win the Meta VSC challenge; subsequent analysis [8] exposes temporal-attack vulnerability. Explicit temporal alignment (VSAL [30], TransVCL [29]) shows order matters for copy localization. We diagnose *when* temporal structure helps versus hurts.

Temporal sensitivity in video models. Lei et al. [40] established the static-bias problem. Subsequent work diagnoses this in *generative* VLM QA [42, 41, 46, 50, 45], including recent hallucination and zero-shot benchmarks [52, 53]; all probe generative outputs, not retrieval embeddings. Arrow-of-time diagnostics [17, 48] and further studies [13, 34, 18, 33, 12] show the failure is structural.

Table 5: Dedicated encoders across all tasks (VCDB: 5,585 pairs; HDD: 1,687 maneuvers; nuScenes: 950 pairs; EPIC: 500 sequences). Nymeria and MUVR scene-matching results replicate the VCDB finding and are in Appendix D. 95% bootstrap CIs in Appendix J.

Method	**VCDB (AP↑)**	**HDD (AP↑)**	**nuScenes (AP↑)**	**EPIC s_{rev}↓**	**EPIC Bal. Acc.↑**
DINOv3 BoF	0.979	0.530	0.613	1.000	N/A
DINOv3 Chamfer	**0.989**	0.559	0.612	0.9999	N/A
DINOv3 Temporal Deriv.	0.775	0.498	0.623	0.500	**1.000**[‡]
DINOv3 Attn. Trajectory	0.700	0.507	0.584	**0.192**	**1.000**[‡]
V-JEPA 2 BoT	0.838	0.825	0.791	0.984	0.500
V-JEPA 2 Encoder-Seq DTW	—	0.942	**0.867**	—	—
V-JEPA 2 Temporal Res.	0.698	**0.956**	0.815	0.699	0.778
ViCLIP (InternVid-10M)	0.907	0.542	0.618	0.996	N/A
X-CLIP (16-frame)	0.960	—	—	0.980	N/A
TARA (chiral, cosine)	—	0.547	—	0.955	N/A
PL-Stitch BoF	—	0.478	—	1.000	N/A
PL-Stitch DTW (raw emb.)	—	0.540	—	0.006	N/A
Optical Flow (RAFT) DTW	—	0.478	—	—	—

— = not evaluated (Encoder-Seq DTW is omitted on VCDB because BoF already saturates at AP = 0.989; the matching stage is not the bottleneck in copy detection). VCDB AP depends on negative seed ($\pm$0.06; Appendix I). [‡]By construction: temporal derivatives negate under reversal, making DTW distance an odd function of frame ordering.

Table 6: VLM embedding probes across all tasks, grouped by pipeline stage. Generative forward/reverse classification is reported separately in Section 3.3.1 (Table 3).

Method	**VCDB (AP↑)**	**HDD (AP↑)**	**nuScenes (AP↑)**	**EPIC s_{rev}↓**	**EPIC Bal. Acc.↑**
Vision tower, pooled					
Gemma-4 (SigLIP)	0.926	0.521	0.662	1.000[†]	N/A
LLaVA (CLIP)	0.915	0.514	0.624	1.000[†]	N/A
Qwen3-VL (SigLIP)	0.491	0.421	0.464	0.998[†]	N/A
Vision tower, per-frame sequence (DTW)					
Gemma-4 (SigLIP)	0.632	0.553	0.469	0.492	**1.000**[‡]
LLaVA (CLIP)	0.637	0.545	0.470	0.495	**1.000**[‡]
Qwen3-VL (SigLIP)	0.610	0.468	0.517	0.542	**1.000**[‡]
LLM backbone, pooled hidden state					
Gemma-4	0.838	0.472	0.569	$\geq$0.998	N/A
LLaVA	0.907	0.512	0.652	$\geq$0.998	N/A
Qwen3-VL	0.916	0.518	0.630	$\geq$0.998	N/A

[†]Pooled representations are order-invariant by construction. [‡]By construction: temporal-derivative DTW is an odd function of frame ordering.

NExT-GQA [39] introduces GQ@τ, a joint grounded-QA metric—complementary to the pair-level AP we report. We differ from all of the above in *domain* (retrieval, where failures are silent) and *granularity* (localizing failure to aggregation via scramble gradients and feature-vs-comparator decomposition). *Training-time* fixes are complementary, not competing: TGPO [35] uses a binary ordered/shuffled RL reward (our gradient extends this to a continuous K-parameterized diagnostic); PL-Stitch [10] and ArrowGEV [11] inject order-sensitivity at pretraining (our decomposition evaluates PL-Stitch and shows order *is* encoded under DTW but does not close the capacity/scale gap); TARA [36] trains with chiral negatives (we use reversal as a probe).

Video prediction for temporal understanding. V-JEPA 2 [1] learns spatiotemporal representations via masked prediction. Drive-JEPA [37] is a downstream application (JEPA distillation for end-to-end driving), not a competing diagnosis; we instead treat V-JEPA 2 as one of several encoders under a retrieval-diagnosis lens. TWLV-I [43] evaluates appearance vs. motion across video foundation models. Li et al. [44] show temporal information is encoded but standard pipelines fail to surface it, matching our finding.

Temporal alignment for retrieval. DTW [31] remains standard; extensions include differentiable DTW [22] and Drop-DTW [49]. At $T \approx 30$ frames, each comparison takes $\sim$1 ms; top-100 rerank-

ing after FAISS [21] adds $\sim$100 ms. We use DTW as a diagnostic tool; Weller et al. [38] provide theoretical justification for multi-vector approaches.

5 Discussion

No single method dominates. **(i) Scene matching** (VCDB, Nymeria, MUVR): BoF wins; temporal methods add noise (Appendix D). **(ii) Motion dynamics** (HDD, nuScenes): per-frame methods and VLM vision towers fail (AP $\leq$ 0.56); the residual advantage grows under sparser sampling (Section 3.2.1). On HDD, 89% of the BoT-to-residual gap traces to the matching stage (per-frame structure with sequence-aware similarity), and on nuScenes the matching stage alone is on par with the residual transform (AP = 0.867 vs. 0.815, marginally overlapping CIs)—within a given encoder, matching, not encoding, is the dominant driver. Video-native ViCLIP [58] ($s_{\text{rev}} = 0.996$, HDD AP = 0.542) and X-CLIP [59] ($s_{\text{rev}} = 0.980$) do not escape the blind spot. Encoders with explicit temporal training fare no better: TARA [36] (HDD AP = 0.547) and PL-Stitch [10] BoF (0.478) fail under cosine; even under DTW, PL-Stitch (0.540, $s_{\text{rev}} = 0.006$—order *is* encoded) trails DINOv3 Chamfer (0.559). V-JEPA 2's spatiotemporal masked-prediction directly supervises motion dynamics; image-level self-supervision (DINOv3) and temporal ranking alone (PL-Stitch) are each insufficient without adequate capacity and pretraining scale. **(iii) Temporal manipulation** (reversal, scrambling): implicit 3D models [9], frame-sampling defenses [8], and Video-LLMs all remain order-invariant [10, 17]. Chiral training (TARA, $s_{\text{rev}} = 0.955$) and a reasoning VLM (Gemini 3.1 Pro, 0.599) improve over baselines but neither reaches reliable. Per-frame features separate forward/reverse under DTW; pooling erases it across all LLM layers.

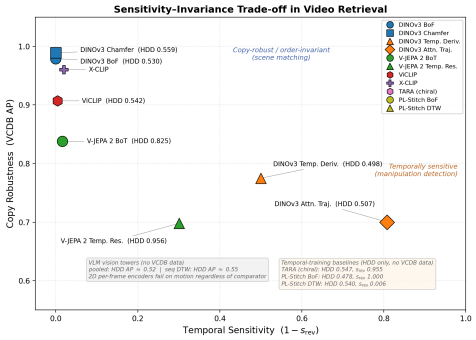

Figure 5: Sensitivity–invariance trade-off: $1 - s_{\text{rev}}$ (EPIC) vs. VCDB AP. No method occupies the upper-right quadrant. **Axis caveat:** pooled and sequence s_{rev} use different similarity functions (Eq. (1)) and are not unit-commensurate; the quadrant claim depends only on within-family rankings.

Limitations. DTW is not indexable ($O(T^2)$); probes train on 400 samples in $p \gg n$ ($D \approx 4{,}000$), bounding recoverability rather than proving absence; ViCLIP's 8-frame input confounds comparison with 64-frame V-JEPA 2 ($\sim$0.20 AP drop at equivalent density, Section 3.2.1). Spatial occlusion attacks (overlays, watermarks, color perturbations) remain a separate, often more prevalent threat orthogonal to temporal blindness.

Practitioner recommendations. (1) Test with the scramble gradient at $K \in \{1, 4, 16\}$: flat AP $\Rightarrow$ temporally blind. (2) For copy detection, high-precision actions (takedowns, bans) favor BoF; soft actions (ranking demotions, review queues) can accept temporal comparators. (3) For motion retrieval, BoF pre-filter $\rightarrow$ top-k DTW rerank adds $\sim$100 ms/query (Appendix Q).

6 Conclusion

No method spans both copy detection and motion retrieval (Figure 5): within an encoder, the matching stage explains 89% of the HDD gap; training-time fixes do not close it. Pooling erases per-frame order signal across VLM layers. Indexable order-sensitive comparators with learnable pooling remain the key open problem.[1]

[1] Code: `https://github.com/mkatindustries/video-temporal-diagnostics`. All datasets and model weights are publicly available; see `REPRODUCIBILITY.md`.

References

[1] M. Assran et al. V-JEPA 2: Self-Supervised Video Models Enable Understanding of Complex Tasks. *arXiv:2506.09985*, 2025.

[2] L. Mur-Labadia et al. V-JEPA 2.1. *arXiv:2603.14482*, 2026.

[3] O. Siméoni et al. DINOv3: Towards Visual Foundation Models with Self-Supervised Learning. *arXiv:2508.10104*, 2025.

[4] Y. Jiang et al. VCDB. In *ECCV*, 2014.

[5] V. Ramanishka et al. Toward driving scene understanding. In *CVPR*, 2018.

[6] H. Caesar et al. nuScenes: A multimodal dataset for autonomous driving. In *CVPR*, 2020.

[7] D. Damen et al. Rescaling egocentric vision: Collection, pipeline and challenges for EPIC-KITCHENS-100. *IJCV*, 2022.

[8] K. Fojcik and P. Syga. Counteracting temporal attacks in video copy detection. *arXiv:2501.11171*, 2025.

[9] R. Deng et al. 3D-CSL: Self-supervised 3D Context Similarity Learning for Near-Duplicate Video Retrieval. *arXiv:2211.05352*, 2022.

[10] C. Che et al. PL-Stitch: Learning Temporal Video Representations via Plackett-Luce Ranking. *arXiv:2511.17805*, 2025.

[11] F. Yu et al. ArrowGEV: Arrow-of-Time Generalized Extreme Value Loss for Video Representation Learning. *arXiv:2601.06559*, 2026.

[12] Y. Zhang et al. WeaveTime: Weaving Temporal Representations for Video Understanding. *arXiv:2602.22142*, 2026.

[13] U. Upadhyay et al. Time Blindness: Why Video-Language Models Can't See What Humans Can? *arXiv:2505.24867*, 2025.

[14] Qwen Team. Qwen3-VL Technical Report. *arXiv:2511.21631*, 2025.

[15] Google DeepMind. Gemma 4 Technical Report. `https://ai.google.dev/gemma/docs/core/model_card_4`, 2026.

[16] Y. Zhang et al. LLaVA-Video: Video Instruction Tuning With Synthetic Data. *arXiv:2410.02713*, 2024.

[17] S. Matta et al. AoT-PsyPhyBENCH: Benchmarking VLMs on arrow-of-time and psychophysical tasks. *arXiv:2510.26241*, 2025.

[18] S. Li et al. VITATECS: A Diagnostic Dataset for Temporal Concept Understanding of Video-Language Models. In *ECCV*, 2024.

[19] Y. Feng et al. Multi-Modal Untrimmed Video Retrieval. *arXiv:2510.21406*, 2025.

[20] A. Abbas et al. SemDeDup: Data-efficient learning at web-scale through semantic deduplication. *arXiv:2303.09540*, 2023.

[21] J. Johnson et al. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 2019.

[22] M. Cuturi and M. Blondel. Soft-DTW: A Differentiable Loss Function for Time-Series. In *ICML*, 2017.

[23] M. Douze, G. Tolias, E. Pizzi et al. The 2021 Image Similarity Dataset and Challenge. In *NeurIPS Competition Track*, 2021.

[24] E. Pizzi, G. Kordopatis-Zilos et al. The 2023 Video Similarity Dataset and Challenge. In *CVPR Workshop*, 2023.

[25] E. Pizzi, S. D. Roy, S. N. Ravindra, P. Goyal, and M. Douze. A Self-Supervised Descriptor for Image Copy Detection. In *CVPR*, 2022.

[26] G. Kordopatis-Zilos, S. Papadopoulos, I. Patras, and I. Kompatsiaris. ViSiL: Fine-grained Spatio-Temporal Video Similarity Learning. In *ICCV*, 2019.

[27] G. Kordopatis-Zilos, C. Tzelepis, S. Papadopoulos, I. Kompatsiaris, and I. Patras. DnS: Distill-and-Select for Efficient and Accurate Video Indexing and Retrieval. *IJCV*, 2022.

[28] S. He et al. A Large-scale Comprehensive Dataset and Copy-overlap Aware Evaluation Protocol for Segment-level Video Copy Detection. In *CVPR*, 2022.

[29] S. He et al. TransVCL: Attention-enhanced Video Copy Localization Network with Flexible Supervision. In *AAAI*, 2023.

[30] Z. Han, X. He, M. Tang, and Y. Lv. Video Similarity and Alignment Learning on Partial Video Copy Detection. In *ACM MM*, 2021.

[31] H. Sakoe and S. Chiba. Dynamic Programming Algorithm Optimization for Spoken Word Recognition. *IEEE Trans. Acoustics, Speech, and Signal Processing*, 26(1):43–49, 1978.

[32] L. Ma et al. Nymeria: A Massive Collection of Multimodal Egocentric Daily Motion in the Wild. In *CVPR*, 2025.

[33] T. V. Sethuraman et al. Stress Tests REVEAL Fragile Temporal and Visual Grounding in Video-Language Models. *arXiv:2602.11244*, 2026.

[34] B. Li et al. TimeBlind: A Spatio-Temporal Compositionality Benchmark for Video LLMs. *arXiv:2602.00288*, 2026.

[35] Z. Xu et al. Incentivizing Temporal-Awareness in Egocentric Video Understanding Models. *arXiv:2603.27184*, 2026.

[36] P. Bagad and A. Zisserman. TARA: Simple and Efficient Time Aware Retrieval Adaptation of MLLMs. *arXiv:2512.13511*, 2025.

[37] L. Wang et al. Drive-JEPA: Video JEPA Meets Multimodal Trajectory Distillation for End-to-End Driving. *arXiv:2601.22032*, 2026.

[38] O. Weller, M. Boratko, I. Naim, and J. Lee. On the Theoretical Limitations of Embedding-Based Retrieval. *arXiv:2508.21038*, 2025.

[39] J. Xiao, A. Yao, Z. Li, and T.-S. Chua. Can I Trust Your Answer? Visually Grounded Video Question Answering. In *CVPR*, 2024. *arXiv:2309.01327*.

[40] J. Lei, T. L. Berg, and M. Bansal. Revealing Single Frame Bias for Video-and-Language Learning. In *ACL*, 2023.

[41] P. Bagad, M. Tapaswi, and C. G. M. Snoek. Test of Time: Instilling Video-Language Models with a Sense of Time. In *CVPR*, 2023.

[42] Y. Liu, S. Li, Y. Liu et al. TempCompass: Do Video LLMs Really Understand Videos? *arXiv:2403.00476*, 2024.

[43] H. Lee, J.-Y. Kim, K. Baek et al. TWLV-I: Analysis and Insights from Holistic Evaluation on Video Foundation Models. *arXiv:2408.11318*, 2024.

[44] L. Li, Y. Liu, L. Yao et al. Temporal Reasoning Transfer from Text to Video. In *ICLR*, 2025.

[45] Z. Shangguan, C. Li, Y. Ding et al. TOMATO: Assessing Visual Temporal Reasoning Capabilities in Multimodal Foundation Models. In *ICLR*, 2025.

[46] M. Cai, R. Tan, J. Zhang et al. TemporalBench: Benchmarking Fine-grained Temporal Understanding for Multimodal Video Models. *arXiv:2410.10818*, 2024.

[47] G. Kordopatis-Zilos, G. Tolias, C. Tzelepis, I. Kompatsiaris, I. Patras, and S. Papadopoulos. Self-Supervised Video Similarity Learning. *arXiv:2304.03378*, 2023.

[48] Z. Xue, M. Luo, and K. Grauman. Seeing the Arrow of Time in Large Multimodal Models. In *NeurIPS*, 2025.

[49] N. Dvornik, I. Hadji, K. G. Derpanis, and A. Garg. Drop-DTW: Aligning Common Signal Between Sequences While Dropping Outliers. In *NeurIPS*, 2021.

[50] B. Feng, Z. Lai, S. Li, Z. Wang, S. Wang, P. Huang, and M. Cao. Breaking Down Video LLM Benchmarks: Knowledge, Spatial Perception, or True Temporal Understanding? *arXiv:2505.14321*, 2025.

[51] T. Wang, F. Ma, Z. Liu, and F. Rao. A Dual-Level Detection Method for Video Copy Detection. *arXiv:2305.12361*, 2023.

[52] J. Yang, Y. Min, J. Zhang, S. Shan, and X. Chen. INFACT: A Diagnostic Benchmark for Induced Faithfulness and Factuality Hallucinations in Video-LLMs. *arXiv:2603.11481*, 2026.

[53] J. Meng et al. VideoZeroBench: Probing the Limits of Video MLLMs with Spatio-Temporal Evidence Verification. *arXiv:2604.01569*, 2026.

[54] Anthropic. Claude 4.6 Opus Model Card. `https://docs.anthropic.com/en/docs/about-claude/models`, 2026.

[55] Google DeepMind. Gemini 3.1 Pro. `https://ai.google.dev/gemini-api/docs/models`, 2026.

[56] Y. Wang, Y. Li, H. Li et al. InternVid: A Large-scale Video-Text Dataset for Multimodal Understanding and Generation. In *ICLR*, 2024.

[57] B. Ni, H. Peng, M. Chen et al. Expanding Language-Image Pretrained Models for General Video Recognition. In *ECCV*, 2022.

[58] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, and A. Smola. Deep Sets. In *NeurIPS*, 2017.

A FPS Cap Invariance Proof

For V-JEPA 2 on Honda HDD, each clip of duration d seconds is sampled at target_fps $= 64/d$ to fill the 64-frame input. An FPS cap only activates when target_fps $>$ cap; otherwise frame selection is identical to the uncapped baseline.

Table 7 confirms that across all 9,811 HDD maneuver segments (at ± 3 s context), the maximum target_fps is 9.14. Caps at 10, 15, and 30 therefore never activate; the identical AP values at these caps are explained by frame selection invariance, not by a learned robustness plateau.

Table 7: FPS cap activation across all HDD segments (± 3 s context). Caps $\geq$10 activate for zero segments, confirming identical frame selection to the uncapped baseline. Unique frames = number of distinct video frames extracted before padding to 64.

FPS Cap	% Active	Mean Unique	Mean Dup%	Effect
2	99.9%	20.2	68.4%	Strong degradation
5	85.9%	49.1	23.3%	Moderate degradation
10	0.0%	64.0	0.0%	No-op (identical to baseline)
15	0.0%	64.0	0.0%	No-op (identical to baseline)
30	0.0%	64.0	0.0%	No-op (identical to baseline)

Segment durations range from 7.0 s to 46.0 s (mean 10.1 s), yielding target_fps $\in [1.39, 9.14]$ (mean 6.66). All values fall below the lowest non-trivial cap (10 FPS), so meaningful degradation occurs only when the cap drops below the native target rate.

B Context Sweep Table

Table 8: Context sweep on HDD (1,687 maneuvers, 64 unique frames, no padding). $\text{fps}_{\text{eff}} = 64 \,/\, d$ where d is clip duration in seconds.

context_sec	**fps_{eff}**	**Temporal Residual AP**	**BoT AP**	**Δ (Res. $-$ BoT)**
1.0	11.1	0.962	0.924	+0.038
2.0	8.1	0.963	0.877	+0.086
3.0	6.4	0.956	0.825	+0.131
4.0	5.3	0.940	0.774	+0.166
6.5	3.8	0.872	0.663	+0.209
8.0	3.2	0.838	0.614	+0.224

Both signals degrade as the window widens, but temporal residual degrades more slowly. The gap grows with sparser sampling ($+0.224$ AP at 8 s): the residual holds up under temporal dilution where spatial content alone does not.

C VLM Layer Ablation

To rule out the possibility that temporal order information is present at different depths of the LLM backbone, we extract hidden states from the last, middle, and penultimate layers of each VLM and compute s_{rev} (Table 9). For comparison, we include vision-token DTW ($s_{\text{rev}} \approx 0.49$), which retains per-frame temporal structure.

Table 9: s_{rev} across LLM layers for three VLM families (EPIC-Kitchens, 500 sequences). Values near 1.0 indicate order-invariance; values near 0.5 indicate perfect forward/reverse separation. Order-invariance persists across all layers in all models.

Model	Vision DTW	LLM Last	LLM Mid	LLM Penult
Qwen3-VL-8B	—*	0.998	0.998	1.000†
Gemma-4-31B	0.492	1.000	1.000†	1.000
LLaVA-Video-7B	0.495	0.998	0.998	0.999

*Qwen3-VL vision DTW omitted due to dynamic-resolution spatial grids (see Table 4). †Clamped from 1.001.

All LLM layers produce $s_{\text{rev}} \approx 1.0$: order-invariance is a property of the full LLM representation under mean-pooled readouts, not an artifact of the final layer. The temporal signal in per-frame vision tokens ($s_{\text{rev}} \approx 0.49$) is not preserved through the vision-to-LLM projection in any pooled summary.

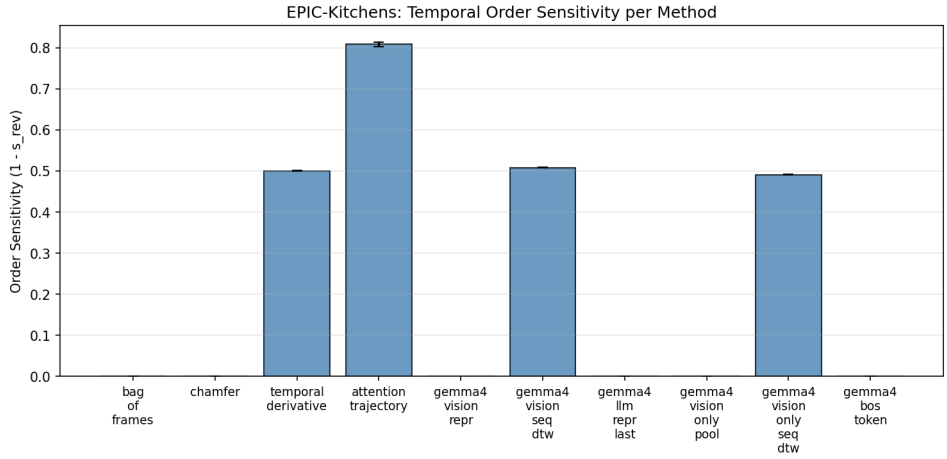

Figure 6: Order sensitivity (s_{rev}) across all embedding probes on EPIC-Kitchens (500 sequences). VLM vision towers are perfectly order-invariant when pooled ($s_{\text{rev}} \approx 1.0$), but per-frame sequences compared via DTW achieve perfect forward/reverse separation (balanced accuracy = 1.0). LLM hidden states remain order-invariant regardless.

D Full Scene Matching Results

We evaluated models across three scene-matching datasets: VCDB [4] (copy detection), Nymeria [32] (egocentric activity discrimination), and MUVR [19] (untrimmed multi-modal news/dance retrieval).

BoF and Chamfer matching consistently dominate. On Nymeria, the 100-session scale confirms that scene appearance remains the strongest signal even with high intra-class variability (AP $\approx$ 0.48). On MUVR News, matching events share visual backgrounds (e.g., news desks), favoring semantic pooling. Attention Trajectories fail because egocentric gaze and semantic pacing vary unpredictably, introducing severe intra-class noise.

Table 10: Retrieval AP driven by scene appearance / visual noise. VCDB: 528 videos, 5,585 positive pairs; Nymeria: 100 egocentric sessions, 8,780 segments; MUVR News: 9,958 videos, 74 topics.

Method	VCDB	Nymeria	MUVR (News)
BoF (DINOv3)	0.979	**0.485**	0.731
Chamfer matching (DINOv3)	**0.989**	0.482	**0.746**
Temporal Residual (V-JEPA 2)	0.698	0.279	0.505
Attention Trajectory (DINOv3)	0.700	0.217	0.466

E Preliminary: Qwen3-VL on VCDB

We first tested whether Video-LLMs can detect temporal reversal. Using Qwen3-VL [14], we presented forward and reversed versions of 49 VCDB videos and asked for temporal direction classification (3 trials each).

Table 11: Qwen3-VL reversal detection on VCDB. Chance = 50%.

Model	Dataset (N)	Fwd Acc.	Rev Acc.	Overall
Qwen3-VL-4B-Instruct	Single clip (1)	100%	0%	50.0%
Qwen3-VL-30B-A3B	VCDB subset (49)	83.0%	28.6%	55.8%

The 30B model achieves 55.8%, near chance. The slight improvement is driven by a content-dependent FORWARD bias (83% forward, 28.6% reverse), not temporal understanding. This motivated the systematic multi-VLM evaluation on EPIC-Kitchens (Section 3.3.1).

F VLM Temporal Integrity Probe

Separately from forward/reverse classification, we asked VLMs to judge whether a video's temporal structure is "INTACT" or "TAMPERED," on forward, reversed, and chunk-scrambled ($K = 4, 16$) versions of EPIC-Kitchens sequences.

Table 12: Temporal integrity probe on EPIC-Kitchens (500 sequences): fraction judged TAMPERED. LLaVA excluded from aggregate due to constant-output degeneracy.

Model	Forward	Reversed	Scramble $K=4$	Scramble $K=16$
Qwen3-VL-8B	0.112	0.870	0.668	0.660
Gemma-4-31B	0.150	0.840	0.202	0.220
LLaVA-Video-7B†	1.000	1.000	1.000	1.000

†LLaVA produces constant "TAMPERED" regardless of input (text-only baseline confirms). Excluded from integrity analysis.

Qwen and Gemma correctly identify forward videos as INTACT ($\sim$85–89%) and both flag reversed videos as TAMPERED (Qwen 87%, Gemma 84%). Scramble detection diverges: Qwen flags 67% of scrambled videos ($K = 4$) as TAMPERED while Gemma flags only 20%, suggesting the two models rely on different heuristics. Cross-transform consistency is very low (3.8–8.2%).

G VLM Prompt Sensitivity and Text-Only Baselines

Answers on forward/reverse classification are dominated by prompt phrasing, not visual content. Gemma-4 exhibits opposite biases across prompts: prompt 0 yields 87% reverse bias; prompt 2 yields 73% forward bias. LLaVA-Video answers "FORWARD" for prompts 0–1 (>98%) but "REVERSE" for prompt 2 (83%). Text-only baselines confirm these are prior biases; Qwen3 shows no text-only bias but still performs at chance with video.

LLaVA-Video produces constant "TAMPERED" regardless of input (text-only baseline confirms prior bias). We exclude it from aggregate integrity statistics in Appendix F.

H HDD Leakage Analysis

Since V-JEPA 2 temporal residuals achieve AP = 0.956 on Honda HDD, well above the next-best method (BoT, AP = 0.825), we verify that no metadata leakage inflates this result.

Feature inputs. All six methods receive *only* decoded video frames. GPS is used exclusively for DBSCAN clustering; timestamps select the frame window (± 3 s) and are discarded before extraction.

Segment duration. Maneuver segments span 7.0–46.0 s (mean 10.1 s). Left and right turns have similar duration distributions (left: 10.3 ± 3.8 s; right: 9.9 ± 3.6 s), ruling out duration as a shortcut. V-JEPA 2 pads all segments to 64 frames, eliminating length signal.

Cluster composition. Only "mixed" clusters containing both turn directions are used. Within each, all $\binom{n}{2}$ pairs are evaluated, ensuring models must distinguish maneuver type at the same location.

Cross-session control. To rule out session adjacency confounds, we re-evaluate excluding same-session pairs (91.4% of pairs are already cross-session). AP is unchanged: V-JEPA 2 temporal residual 0.957 (cross-session) vs. 0.956 (all pairs); BoF 0.536 vs. 0.530. This confirms results are not inflated by within-session visual correlations.

Mechanistic explanation. The context sweep (Section 3.2.1) shows temporal residual AP degrades monotonically as context narrows, consistent with genuine temporal processing. BoF methods remain near chance across all windows.

Motion proxy check. To rule out trivial motion magnitude as a shortcut, we compute mean pixel difference between consecutive frames for each segment and correlate pairwise proxy similarity with V-JEPA 2 residual similarity across all 97,731 pairs. Spearman $r = 0.051$; the motion proxy itself achieves only AP = 0.474 (below chance). Residuals capture motion *structure*, not intensity.

I VCDB Negative-Sampling Sensitivity

Our primary VCDB evaluation uses 1:1 positive-to-negative sampling (5,585 pairs each). The full dataset contains $\binom{528}{2} \approx 139$K possible pairs. We re-evaluate at three ratios to verify robustness (Table 13).

Table 13: VCDB AP across negative-sampling ratios. *Rank order and relative gaps are stable across all settings.* All-pairs uses the full $\binom{528}{2}$ set (requiring $\sim$140K DTW comparisons).

Method	1:1	1:5	All pairs
DINOv3 Chamfer	0.989	0.949	0.845
DINOv3 BoF	0.979	0.929	0.829
DINOv3 Temporal Deriv.	0.775	0.424	0.228
DINOv3 Attn. Trajectory	0.700	0.408	0.225

Rank order (Chamfer > BoF > Temporal Deriv. > Attn. Trajectory) is preserved across all ratios. AP decreases with more negatives as expected; AUC remains stable (e.g. Chamfer AUC: 0.990, 0.989, 0.989).

J Bootstrap Confidence Intervals

Table 14 reports 95% bootstrap confidence intervals (2,000 resamples, percentile method, seed = 42) for the headline AP values from Tables 1–3. Degenerate bootstrap samples (all-positive or all-negative) fall back to the point estimate.

Table 14: 95% bootstrap CIs for AP across all benchmarks. "Width" is $\text{CI}_{\text{high}} - \text{CI}_{\text{low}}$. VCDB and HDD CIs are narrow (<0.02); nuScenes, Nymeria, and MUVR CIs are wider due to smaller evaluation sets.

Benchmark	Method	AP	95% CI	Width
VCDB	DINOv3 Chamfer	0.989	[0.987, 0.991]	0.004
	DINOv3 BoF	0.979	[0.976, 0.981]	0.005
	DINOv3 Temporal Deriv.	0.775	[0.765, 0.784]	0.019
	DINOv3 Attn. Trajectory	0.700	[0.689, 0.711]	0.022
	V-JEPA 2 BoT	0.838	[0.829, 0.846]	0.017
	V-JEPA 2 Temporal Res.	0.698	[0.687, 0.709]	0.022
HDD	V-JEPA 2 Temporal Res.	0.956	[0.955, 0.958]	0.003
	V-JEPA 2 BoT	0.825	[0.821, 0.828]	0.007
	DINOv3 Chamfer	0.559	[0.555, 0.564]	0.009
	DINOv3 BoF	0.529	[0.525, 0.534]	0.009
	DINOv3 Attn. Trajectory	0.507	[0.502, 0.512]	0.010
	DINOv3 Temporal Deriv.	0.498	[0.494, 0.503]	0.009
nuScenes	V-JEPA 2 Encoder-Seq DTW	0.867	[0.842, 0.890]	0.048
	V-JEPA 2 Temporal Res.	0.815	[0.785, 0.844]	0.059
	V-JEPA 2 BoT	0.791	[0.761, 0.820]	0.059
	DINOv3 Temporal Deriv.	0.623	[0.580, 0.665]	0.085
	DINOv3 BoF	0.613	[0.570, 0.658]	0.088
	DINOv3 Chamfer	0.612	[0.570, 0.659]	0.089
	DINOv3 Attn. Trajectory	0.584	[0.540, 0.634]	0.094
Nymeria	DINOv3 BoF	0.485	[0.482, 0.489]	0.007
	DINOv3 Chamfer	0.482	[0.478, 0.485]	0.007
	DINOv3 Temporal Deriv.	0.274	[0.271, 0.277]	0.006
	DINOv3 Attn. Trajectory	0.217	[0.215, 0.219]	0.004
	V-JEPA 2 BoT	0.420	[0.417, 0.424]	0.007
	V-JEPA 2 Temporal Res.	0.278	[0.276, 0.281]	0.005
MUVR News	DINOv3 BoF	0.731	[0.724, 0.740]	0.016
	DINOv3 Chamfer	0.745	[0.738, 0.754]	0.016
	DINOv3 Temporal Deriv.	0.476	[0.468, 0.485]	0.017
	DINOv3 Attn. Trajectory	0.466	[0.458, 0.475]	0.017
	V-JEPA 2 BoT	0.597	[0.587, 0.605]	0.018
	V-JEPA 2 Temporal Res.	0.505	[0.496, 0.514]	0.018

VCDB and HDD CIs are narrow (<0.025) under pair-level resampling. nuScenes CIs are wider (0.06–0.09) due to the smaller evaluation set (950 pairs), but rank order remains unambiguous. Nymeria CIs are very tight (<0.01) despite the scene-retrieval setting (394,210 pairs). MUVR News CIs (0.016–0.018) reflect the moderate evaluation set (27,453 pairs). **Note:** pair-level resampling underestimates CI width because pairs within a GPS cluster share clips. A cluster-level block bootstrap (resampling clusters with replacement) produces CIs 7–32× wider on HDD: e.g., V-JEPA 2 temporal residual widens from $[0.955, 0.958]$ to $[0.890, 0.978]$. Rank order is preserved; the pair-level CIs should be read as lower bounds on uncertainty.

K DTW Similarity-Transform α-Invariance

DTW similarities are computed as $s = \exp(-\alpha \cdot d)$ where d is the (path-normalised) DTW distance and α controls the sharpness of the monotone transform. Since α applies a strictly monotone transformation to all pairwise distances, it cannot change ranking and therefore cannot change AP. We verify this empirically on three representative cells (Table 15).

L HDD Qualitative Analysis

Complementing the quantitative results, we present qualitative evidence that V-JEPA 2 temporal residuals capture genuine motion dynamics rather than a visual shortcut.

Table 15: AP across $\alpha \in \{0.5, 1, 2, 5, 10\}$. All cells show spread $= 0.000$ (AP identical to three decimal places), confirming α-invariance within numerical precision.

Cell	$\boldsymbol{\alpha = 0.5}$	$\boldsymbol{\alpha = 1}$	$\boldsymbol{\alpha = 2}$	$\boldsymbol{\alpha = 5}$	$\boldsymbol{\alpha = 10}$
DINOv3 HDD Temporal Deriv.	0.476	0.476	0.476	0.476	0.476
V-JEPA 2 HDD Encoder-Seq	0.942	0.942	0.942	0.942	0.942
DINOv3 VCDB Temporal Deriv.	0.775	0.775	0.775	0.775	0.775

DINOv3 HDD Temporal Deriv. AP (0.476) differs from Table 5 (0.498) because the sweep uses cached features extracted with different preprocessing; rank order and α-invariance are unaffected. 95% bootstrap CIs: DINOv3 HDD [0.471, 0.480]; V-JEPA 2 HDD [0.940, 0.943]; DINOv3 VCDB [0.765, 0.784].

Nearest-neighbor examples. For three representative intersection clusters, the top-3 neighbors by temporal residual similarity retrieve the same maneuver type despite identical backgrounds; errors occur when both turns share similar speeds through the same intersection phase.

Background similarity statistics. Within each GPS cluster, all segments share the same intersection. BoF cosine similarity for all $\binom{n}{2}$ pairs per cluster: mean 0.63 ± 0.18 (same-direction: 0.63 ± 0.18; cross-direction: 0.60 ± 0.16; gap = 0.029), confirming appearance is non-discriminative. V-JEPA 2 temporal residual similarity aligns with maneuver labels (same: 0.008 ± 0.006; cross: 0.003 ± 0.001). Figure 7 shows the score distributions.

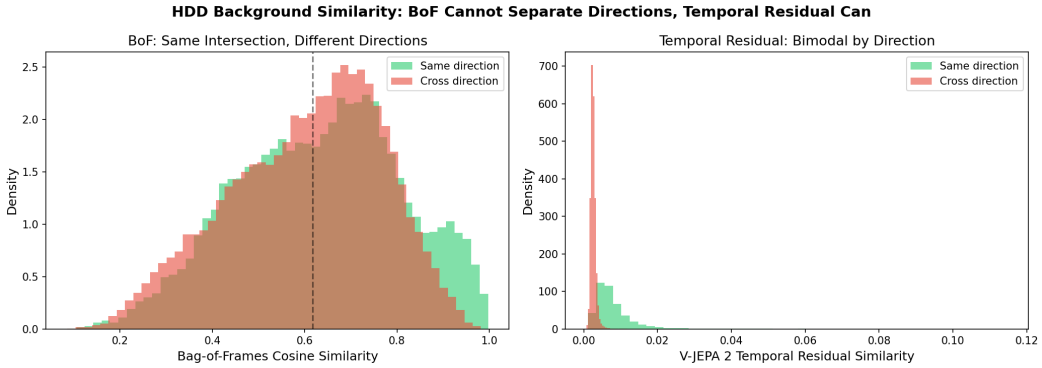

Figure 7: Pairwise similarity distributions on HDD intersection clusters. BoF similarities overlap heavily regardless of maneuver match, while V-JEPA 2 temporal residual scores separate same- vs. cross-direction pairs.

M nuScenes Cross-Dataset Visualization

N LLM Hidden-State Probe: Vision-Token-Only Analysis

A potential objection to our LLM hidden-state probe is that mean-pooling over *all* tokens reintroduces symmetric aggregation. To address this, we extract LLM hidden states restricted to *vision-token positions only*. Three variants:

1. **Vision-only mean pool**: Mean-pool LLM hidden states at vision-token positions only. Compare via cosine similarity.
2. **Vision-only per-frame sequence**: Group vision-token hidden states by their source frame, pool per frame to obtain a (T, D) sequence. Compare via temporal derivative DTW.
3. **BOS/first token**: Use the first token's hidden state (BOS or system token) as a summary representation. Compare via cosine similarity.

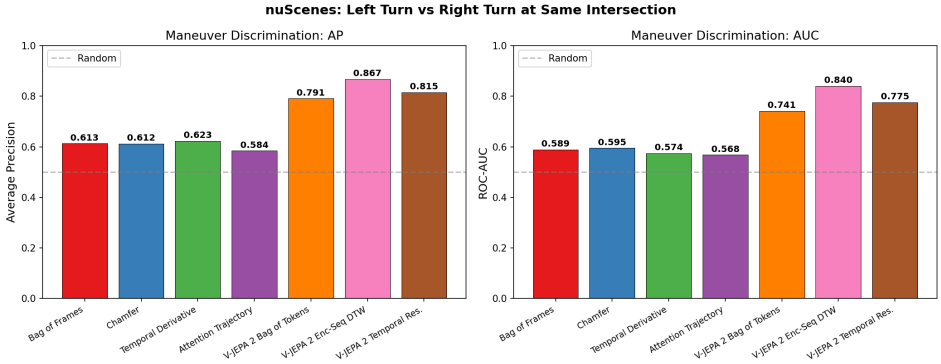

Figure 8: nuScenes cross-dataset validation (264 segments, 950 pairs, 50 intersection clusters). V-JEPA 2 encoder-sequence DTW (AP = 0.867) is the strongest method; the matching-stage change alone is on par with the temporal residual (AP = 0.815; CIs marginally overlap, Appendix J).

Table 16: s_{rev} for vision-token-only LLM hidden-state probes (EPIC-Kitchens, 500 sequences). If vision-only pooling recovers temporal signal, we expect $s_{\text{rev}} \ll 1.0$; if order-invariance persists, the conclusion from Section 3.3.2 is strengthened.

Model	**Vision-only pool**	**Vision-only seq DTW**	**BOS token**
Qwen3-VL-8B	0.998 [0.998, 0.998]	**0.542** [0.540, 0.544]	0.996
Gemma-4-31B	1.000 [1.000, 1.000]†	**0.509** [0.509, 0.509]	1.000
LLaVA-Video-7B	0.999 [0.998, 0.999]	**0.517** [0.516, 0.518]	1.000†

Vision-only mean pooling remains $s_{\text{rev}} \approx 1.0$ across all models, confirming that symmetric aggregation, not the LLM backbone, erases temporal signal. The per-frame sequence probe yields low s_{rev} (balanced accuracy = 1.0), showing that LLM hidden states preserve per-position order information that pooling discards. BOS tokens are order-invariant, indicating the model's summary position captures no temporal signal. †Raw values >1.0 from f32 precision; clamped to 1.000.

O Scramble Gradient Data

V-JEPA 2 residuals show non-monotonic degradation (e.g. $K = 3$ higher than $K = 2$), likely due to boundary effects where the spatiotemporal prediction window crosses chunk boundaries. A multi-seed analysis (10 independent shuffle seeds per K) shows that order-invariant methods have zero variance ($\sigma = 0.000$), while temporal methods show small but real variation ($\sigma \leq 0.007$) that does not affect method ranking (Figure 9). Because the main scramble gradient shuffles *extracted* embeddings, it tests whether the comparator (DTW) is sensitive to temporal order, not whether V-JEPA's encoder responds differently to scrambled input. A raw-frame scramble (re-running V-JEPA 2 on chunk-shuffled video) produces clean monotonic degradation: residual AP drops from 0.705 at $K = 1$ to 0.652 at $K = 16$, while BoT remains flat at 0.839. The encoder does produce different residuals from temporally disrupted input.

P Linear Probe on LLM Hidden States

The s_{rev} analysis (Section 3.3.2) shows that mean-pooled LLM hidden states are order-invariant. A natural follow-up: is temporal signal present in the per-position hidden states but lost to symmetric aggregation? If so, a learned classifier should recover it.

We extract last-layer hidden states from all three VLMs for 200 EPIC-Kitchens sequences (forward and reverse), yielding 400 samples per model. We train ℓ_2-regularized logistic regression (5-fold GroupKFold CV, fwd/rev pairs kept together, $C \in \{0.01, 0.1, 1, 10, 100\}$) on seven readout strategies:

- **mean**: mean over all positions (replicates s_{rev} finding)

Table 17: AP across scramble levels K on VCDB (527 of 528 videos; one decode failure). $K = 1$ is the unscrambled baseline. Order-invariant methods show constant AP; sequence-aware methods show overall degradation (multi-seed analysis: $\sigma \leq 0.007$; Figure 9).

Method	$K=1$	$K=2$	$K=3$	$K=4$	$K=6$	$K=8$	$K=12$	$K=16$
BoF (DINOv3)	0.979	0.979	0.979	0.979	0.979	0.979	0.979	0.979
Chamfer (DINOv3)	0.989	0.989	0.989	0.989	0.989	0.989	0.989	0.989
V-JEPA 2 BoT	0.838	0.838	0.838	0.838	0.838	0.838	0.838	0.838
Temporal Derivative (DINOv3)	0.775	0.735	0.733	0.734	0.724	0.731	0.729	0.722
V-JEPA 2 Temporal Residual	0.698	0.612	0.633	0.609	0.601	0.618	0.609	0.639
Attention Trajectory (DINOv3)	0.700	0.650	0.631	0.619	0.598	0.596	0.585	0.572

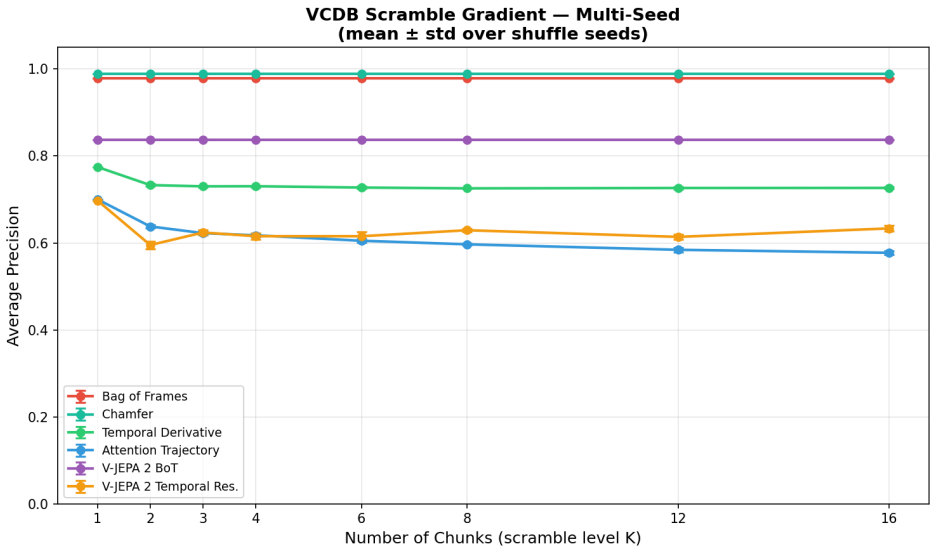

Figure 9: Multi-seed scramble gradient on VCDB (10 seeds per K, mean $\pm$ std shaded). Order-invariant methods (BoF, Chamfer, V-JEPA 2 BoT) have zero variance. Temporal methods show small variance ($\sigma \leq 0.007$) that does not change the ranking.

- **max**: element-wise max over all positions
- **first / last**: single position (BOS or final token)
- **concat_first_last**: concatenation of first and last positions
- **vision_mean**: mean restricted to vision-token positions
- **vision_concat_fl**: first and last vision-token positions

Neither linear nor nonlinear probes recover temporal order from LLM hidden states. All 126 configurations (3 models $\times$ 7 pooling strategies $\times$ {5 regularization strengths for linear, 1 for MLP}) land between 0.458 and 0.560, within the range expected from chance on 400 samples. The "first" token gives exactly 0.500 under both probes, a sanity check that the pipeline is correct when the input is truly identical across conditions (BOS tokens do not change between forward and reverse).

Note that per-position LLM hidden states *do* carry order signal when compared as a sequence via DTW ($s_{\text{rev}} \approx 0.51$–$0.54$; Table 16): order is encoded in the relative content across positions, but no fixed-dimensional reduction of the sequence retains it. This is the same pooling-destroys-order dynamic observed at the vision encoder level, replicated one layer deeper.

Q Computational Costs

Table 19 summarizes per-video extraction and per-pair comparison costs on a single H200 GPU. The two-stage retrieval pipeline recommended in Section 5—BoF cosine pre-filter followed by DTW

Table 18: Probe accuracy (GroupKFold CV, fwd/rev pairs kept together) for forward vs. reverse classification from LLM hidden states. Linear = logistic regression (best across $C \in \{0.01, 0.1, 1, 10, 100\}$); MLP = 2-layer, 256 hidden units, ReLU. Chance = 0.500. No configuration exceeds chance.

	Gemma-4 31B		**LLaVA-Video 7B**		**Qwen3-VL 8B**	
Pooling	**Linear**	**MLP**	**Linear**	**MLP**	**Linear**	**MLP**
mean	0.512	0.552	0.500	0.505	0.537	0.510
max	0.532	0.458	0.470	0.490	0.528	0.502
first	0.500	0.500	0.500	0.500	0.500	0.500
last	0.510	0.507	0.515	0.495	0.550	0.512
concat_first_last	0.515	0.490	0.505	0.478	0.555	0.520
vision_mean	0.520	0.550	0.502	0.512	0.537	0.505
vision_concat_fl	0.528	0.500	0.495	0.467	0.515	0.535

reranking on the top-k—adds $\sim$100 ms overhead per query at $k = 100$, dominated by DTW on derivative or residual sequences. Feature extraction is the one-time cost per video at index build; comparison cost governs query latency.

Table 19: Per-video feature extraction and per-pair comparison costs on a single H200 GPU. T = number of frames per clip.

Method	**Extraction (ms/video)**	**Comparison (ms/pair)**
DINOv3 BoF ($T = 30$)	$\sim$120	$<$0.01 (cosine)
DINOv3 Chamfer ($T = 30$)	$\sim$120	$\sim$0.3 (frame matrix)
DINOv3 Attn. Trajectory ($T = 30$)	$\sim$150	$\sim$1 (DTW)
DINOv3 Temporal Deriv. ($T = 30$)	$\sim$120	$\sim$1 (DTW)
V-JEPA 2 BoT ($T = 64$)	$\sim$280	$<$0.01 (cosine)
V-JEPA 2 Temporal Res. ($T = 64$)	$\sim$560	$\sim$1 (DTW)
VLM vision pooled	$\sim$800–2,500	$<$0.01 (cosine)
VLM LLM hidden state	$\sim$1,500–5,000	$<$0.01 (cosine)

V-JEPA 2 temporal residual requires both encoder and predictor forward passes, roughly doubling extraction cost vs. BoT. DTW comparison at $T \approx 30$ is $\sim$1 ms; a top-100 reranking pass after FAISS [21] pre-filtering adds $\sim$100 ms total, making two-stage retrieval practical for production workloads.

R Dataset and Model Licenses

All experiments use publicly available datasets and pretrained models. Table 20 lists the governing license for each asset. Use is restricted to non-commercial research, consistent with the most restrictive license among the assets combined in any given experiment. No new datasets or model weights are released by this paper.

Honda HDD access required a signed Data Use Agreement with HRI-USA; nuScenes required account registration at nuscenes.org; other datasets are downloadable under their respective non-commercial licenses. All assets were used for non-commercial research evaluation only.

Table 20: Licenses for datasets and pretrained model weights used in this work. Verbatim license text is available from each asset's canonical distribution channel; see `REPRODUCIBILITY.md` for HuggingFace IDs and download URLs.

Asset	**License**	**Distribution**
Datasets		
VCDB [4]	Research-only; no warranty (Fudan Univ.)	Project page
Honda HDD [5]	HRI-USA Data Use Agreement (non-commercial)	HRI-USA
nuScenes [6]	CC BY-NC-SA 4.0	nuscenes.org
EPIC-Kitchens-100 [7]	CC BY-NC 4.0	epic-kitchens.github.io
Nymeria [32]	CC BY-NC 4.0	facebookresearch GH
MUVR [19]	Research-only (authors' terms)	Authors' release
Pretrained model weights		
DINOv3 [3]	Meta FAIR non-commercial	HuggingFace
V-JEPA 2 [1]	CC BY-NC 4.0	HuggingFace
ViCLIP / InternVid [58]	Apache 2.0 (code); CC BY-NC 4.0 (weights)	OpenGVLab
X-CLIP [59]	MIT	microsoft/VideoX
Qwen3-VL-8B [14]	Apache 2.0	HuggingFace
Gemma 4 [15]	Gemma Terms of Use	HuggingFace
LLaVA-Video [16]	Apache 2.0 (code); Llama-2 terms (weights)	HuggingFace
RAFT (optical flow)	BSD-3-Clause	princeton-vl/RAFT